PROGRAMMING IN JAVA - EVALUATED COURSE PROJECT

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Course : *Programming in Java*

Course Code : *CSE2006*

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1. Executive Summary

The Personal Study Manager is a console-based Java application designed to help users track their educational materials. It allows users to log different types of study resources—specifically books and video lectures—and monitor their overall completion progress. The application features persistent data storage, ensuring that the user's study log is saved between sessions.

2. Core Features

- **Resource Logging :** Users can add new books (tracking author and page count) and video lectures (tracking URL and duration).
- **Progress Tracking :** The system calculates and displays the user's overall completion percentage based on finished vs. unfinished resources.
- **Search Functionality :** Users can filter their library by entering a specific subject.
- **State Management:** Users can update the status of any resource from "In Progress/To Watch" to "Finished/Watched".
- **Data Persistence :** The application automatically saves the user's library to a local file upon exit and reloads it upon startup.
- **Factory Reset :** A built-in feature allows users to wipe all in-memory data and delete the physical save file.

3. Object-Oriented Architecture

The application makes strong use of core Object-Oriented Programming (OOP) principles to keep the code modular and scalable:

- **Abstraction :** The StudyResource class is abstract, meaning it cannot be instantiated directly. It serves as a blueprint for all future resource types, enforcing a standard structure.
- **Inheritance :** Both Book and VideoLecture inherit common attributes (title, subject, completion status) from the StudyResource parent class, reducing code duplication.
- **Polymorphism :** The displayDetails() method is abstract in the parent class and uniquely overridden in each child class. This allows the StudyManager to loop through a mixed list of resources and call the correct display format for each item seamlessly.
- **Encapsulation :** Class fields (like title, author, durationMinutes) are marked as private and accessed only through designated getter methods, protecting the data from unwanted external modification.

4. Technical Specifications

Component	Implementation Details
Language	Java (Standard Edition)
Data Structure	ArrayList<StudyResource> for dynamic, in-memory data storage.
User Input	java.util.Scanner handles interactive console inputs.
Data Persistence	java.io.Serializable combined with ObjectOutputStream and ObjectInputStream.
Storage Format	Binary serialized file (study_data.ser).
Error Handling	try-catch blocks manage file I/O exceptions during saving and loading.

5. Code

- Code is provided in the repository (given through link)

6. Output Results

- Main menu :

```
=== Welcome to Your Personal Study Manager ===

--- MAIN MENU ---
1. Add a Book
2. Add a Video Lecture
3. View All Resources & Progress
4. Search by Subject
5. Mark Resource as Completed
6. Reset all data
7. Exit
Choice :
```

- Choice 1 :

```
Choice : 1
Enter Title : Java Programming
Enter Subject : Java
Enter Author : ABC
Enter Total Pages : 1100
Book added !!
```

- Choice 2 :

```
Choice : 2
Enter Title : Transform Techniques
Enter Subject : Mathematics
Enter URL : https://youtu.be/EDVJotmT584?si=kiJnExwAkRVPp_r8
Enter Duration : 10
Video added !!
```

- Choice 3 :

```
Choice : 3

--- Your Study Resources ---
[Book] Java Programming | Subject : Java
Author : ABC | Pages : 1100
Status : In Progress
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[VIDEO] Transform Techniques | Subject : Mathematics
URL : https://youtu.be/EDVJotmT584?si=kiJnExwAkRVPp_r8 | Duration : 10 mins
Status : To Watch
--------------------------
OVERALL PROGRESS : 0.00%
--------------------------
```

- Choice 4 :

```
Choice : 4
Enter subject to search : Mathematics
[VIDEO] Transform Techniques | Subject : Mathematics
URL : https://youtu.be/EDVJotmT584?si=kiJnExwAkRVPp_r8 | Duration : 10 mins
Status : To Watch
--------------------------
```

- Choice 5 **:**

```
Choice : 5
Enter the Title of the resource you finished : Java Programming
Status updated for : Java Programming
```

- Choice 3 (again) **:**

```
Choice : 3

--- Your Study Resources ---
[Book] Java Programming | Subject : Java
Author : ABC | Pages : 1100
Status : Finished
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[VIDEO] Transform Techniques | Subject : Mathematics
URL : https://youtu.be/EDVJotmT584?si=kiJnExwAkRVPp_r8 | Duration : 10 mins
Status : To Watch
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OVERALL PROGRESS : 50.00%
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```

- Choice 7 **:**

```
Choice : 7
Data saved. Happy Studying! Goodbye.
```

The data will be automatically saved in a file named 'study_data.ser' upon exiting the console.

7. Recommended Future Enhancements

- **Input Validation:** Add checks to ensure users don't enter negative numbers for pages/durations, or empty strings for titles.
- **Data Format Migration:** Transition from Java Serialization (.ser) to JSON or CSV. This makes the data human-readable and easier to export to other applications.
- **Unique Identifiers (IDs):** Currently, marking a resource as complete relies on the exact title. Adding a unique integer ID to each resource would make searching and updating much more robust.
- **Sorting Capabilities:** Allow the user to sort their list alphabetically by title, or group items by subject before displaying.